



The message hidden within the pattern: a reverse alignment problem for debates in artificial intelligence

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Abstract

This paper explores what I call the reverse alignment problem (RAP). The alignment problem in artificial intelligence (AI) is the challenge of ensuring that superintelligent AI systems harmonize and promote wider social, personal, and environmental values. Standard formulations of the alignment problem depict the issue largely as a technical or engineering problem, one that concerns the right specification of the goals and objectives to be pursued to ensure machines are consistent with the intentions of the designer. This paper builds on this literature by exploring a bidirectional interaction between human values and the machines we use to effectuate our intentions. I argue that alignment works in both ways: we could align machines with the complex tapestry of human values (notoriously difficult); or we can reduce and simplify human values, preferences, and goals so as to be easier to satisfy—a process that is, or so I argue, already underway. Thus, the RAP identifies the way in which we are already paving the way for forms of alignment that may not be desirable, indeed, may be misaligned with the social values for which they were initially intended. To explore this interaction in depth, I look toward the notion of value capture, as formulated by C.T. Nguyen, as the foundational mechanism through which the reverse alignment operates. Value capture occurs when rich, multidimensional human values are reduced to simplified proxies that optimization systems can measure and maximize. These values are then fed back into society and often tacitly adopted, leading to further alignment with machine-readable interpretations of human behavior. As philosopher of technology Karen Crawford has observed on emotion recognition in machine learning, “Techniques have been developed to reduce the messiness of feelings, interior states, preferences, and identifications into something quantitative, detectable, and trackable” (Crawford 2021). Building on this insight, I argue that the RAP is the process in which the human dimension of the alignment problem is surreptitiously modified, modulated, and manipulated to align more easily with machines: alignment achieved in the reverse direction—not by making machines more like humans and, therefore, sensitive to contextual features of the value landscape, but by making humans more ‘machine-like’.

Keywords Alignment problem · Values · Value capture · Artificial intelligence · Reverse alignment problem

1 Introduction

“The analogy between people and machines is pretty exact”

—Jerry Fodor, *Language of Thought* (1975: 125)

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We live, by many accounts, in a time of a great flattening. Values are being flattened (Nguyen 2024); culture is being flattened (Chayka 2024); language is being flattened (Agarwal et al. 2025; Turk 2025). More generally, behavior is being flattened: compressed into clean data flows that continuously and continually sustain large-scale computer infrastructures (Zuboff 2019). In other words, the technological transformations are so momentous and significant that our very way of being in, and relating to, the world is becoming distorted, morphed, skewed, shifted—in a word, *flattened*. Kate Crawford writes specifically about an ‘epistemological violence’ (2021: 221) that large-scale artificial intelligence (AI) industries encourage in formatting the world into a machine-legible order. As she writes, “This epistemological

flattening of complexity into clean signals for the purpose of prediction is now a central logic of machine learning” (2021: 213). What underlies these varying flattening processes, such as they are, is the incorporation of digital technologies into the capillaries of everyday life: from the always-online and operating constellation of smart technologies mediating our lives and gaming algorithms modifying our news feed, to the increasingly personalized incorporation of large language models (LLMs) and various AI agents utilized in the name of productivity and efficiency.

Concurrently, we are also living in a time when researchers endeavor to align AI technologies to human values. This is called the *value alignment problem* (Christian 2020), which, plainly stated, is the challenge of ensuring that AI systems pursue goals that match human values or interests rather than unintended or undesirable goals (Ngo et al. 2025). The challenge is so prodigious that it is called the “most important problem for humanity to ever solve” (Tegmark 2018) and warrants entire institutes to promote the danger of misaligned AI with attention-grabbing book titles like *If Anyone Builds It, Everyone Dies* (Yudkowsky & Soares 2025). With such wide-ranging implications (as the attendant terminology ‘catastrophic’ and ‘existential’ risk implies), it is surprising to see a lack of diversity represented in these debates (cf. Buolamwini 2024). This is due, in part, to the value alignment problem being predominantly configured as a mathematical, technical, or engineering challenge (hence the conversation being reserved for a rather privileged set of individuals in a rather privileged area of the world). On this understanding, the alignment problem concerns the right algorithmic specification of the goals or objectives to be pursued to ensure that the machine pursuing these goals aligns with the designer’s intentions. While few scholars explicitly assert stable, universal values to which machines should be aligned, the locus of alignment is placed squarely on machines: AI aligning with the complex value landscape in which humans live.

On reflection, though, this is a remarkable state of affairs: human values, culture, language, behavior, and meaning-making are all—allegedly—being flattened; and yet we are trying to align AI systems with (ostensibly flattened) values, culture, language, behavior, and meaning-making. As it happens, or so I will argue, this is far from a coincidence. In this paper, I reframe the debate on value alignment through the lens of what I call the *reverse alignment problem* (hereafter, the RAP). The RAP, I argue, expresses the simultaneous and seemingly intertwined engines of value flattening and value alignment that operate through an overlapping and interconnected network of mechanisms, infrastructures, and institutions that facilitate a simplification and, at times, homogenization of the human value landscape, rendering human values more ‘align-able’ for machines. It is indeed an intellectual curiosity that a time so proud and optimistic

of technological progress (‘feel the AGI’, as Ilya Sutskever was known to enthuse to employees at OpenAI) could coincide with the progress-shattering sentiment of a great flattening. It is all the more remarkable that the connections which make this more than a coincidence are rarely explicitly elaborated in alignment research. However, there is a growing field of literature called *sociotechnical alignment* (Edelman et al. 2025) that redresses the situation by framing the value alignment problem as not strictly a question of engineering but equally as one of social and philosophical importance. In proposing the RAP, I seek to contribute to this growing field on sociotechnical alignment.

Sociotechnical alignment and the RAP are concerned with the well-documented instance of strong (often coercive) economic and fiscal incentives that encourage humans to adapt to emerging digital technologies (Del Vicario et al. 2016; Nguyen 2024; Kazienko and Cambria 2024; Pellegrino & Stasi 2024). In other words, they focus less on (or in addition to) autonomous or rogue AI scenarios and shift the focus instead to loci of power and agency operating prior to, during, and after the development of advanced autonomous AI systems. This is important because debates on alignment tend to view problems of power and agency as emerging somewhat suddenly and spontaneously along with autonomous AI systems—like Athena springing fully formed from the head of Zeus. This overlooks what Kulveit and colleagues call ‘gradual disempowerments’ Kulveit et al. (2025) that lead to incremental and sometimes surreptitious loss of human control over societal outcomes that themselves introduce power asymmetries. These asymmetries establish a further contextual layer in which (mis)aligned AI technologies can be established.

The RAP, then, adds to the growing body of literature that identifies power structures that are themselves misaligned. If the systems in which eventual autonomous AI systems are misaligned, then there is little a priori reason to think autonomous systems, having sprung from these misaligned infrastructures, would diverge from the operationalized logic of their parent institutions. As Edelman et al. write, “AI does not exist in a vacuum; they are embedded within larger institutions like companies, states, professional bodies, and therefore beneficial societal outcomes cannot be guaranteed by aligning individual AI systems with their operator’s or user’s intentions” (2025: 1). A large language model (LLM), for instance, might be locally aligned with its developer (OpenAI, Anthropic), but if the stewards of this technology are themselves misaligned with wider social considerations, then the alignment problem is not ‘solved’. One implication of this is that alignment *simpliciter* does not provide enough normative guidance on how alignment should be achieved—a broader discussion is needed that requires expanding the discourse beyond a computational, technical, or engineering lens.

Before beginning, I want to fold the foregoing discussion into a more philosophical consequence of the RAP. Framed as such, the RAP might be seen as sympathetic to a pedigreed line of philosophical research that identifies inherent limitations to machine architectures and computers which prevent them from ever emulating complex features associated with human mentation. Representative of this line of thinking is, *inter alia*, John Searle’s monumental paper (1980), ‘Minds, Brains and Programs’, in which he introduces the Chinese room thought experiment; and Hubert Dreyfus’s provocative (at the time) book, *What Computers Still Can’t Do* (1972). The arguments here are nuanced but tend to revolve around a central conceptual axis: that of identifying unique characteristics of human or biological minds, delineating the contrasts with computers, and concluding that the latter can never be more than an ersatz imitation of the former. In light of current advancements in AI, especially work on LLMs, Dreyfus’s and Searle’s arguments have not aged well—with fully conversational chatbots far exceeding what we thought was possible from a ‘stochastic parrot’ (Bender and Koller 2020). The RAP, then, might seem destined for a similar fate. However, I do not depend on these arguments, or this argumentative strategy, to establish the RAP herein. Rather, if we take the above remarks seriously, the argument I am making is not that contemporary AIs or LLMs will *never* be complex enough to accommodate, adapt to, or engage the textured tapestry of the human value landscape. The argument is that *they might never need to*. Through gradual disempowerments, misaligned economic power structures, and insidious modifications and manipulations of human experience that form the backbone of large-scale, planetary technological infrastructures, we do not have to necessarily make computers complex enough to actually emulate the human mind. The RAP is premised on the assumption, already implicit in major technology companies, that it is comparatively easier to transform human values, goals, and behavior into a machine-legible format than it is to complexify machine architectures to embody the qualitative, vacillatory, unfixed, locally specific, and ambiguous nature of human decision-making, practical reasoning, and value deliberations (Birhane 2021). Situated in relation to the preceding discussion, the RAP frames alignment as a question of being achieved either through making machines more sensitive to the human value landscape or through transforming human mental life and human values into a form more easily legible, amenable, and modifiable for machines—alignment achieved in the reverse direction.¹

¹ Of course, one might ask: why not both? It is true that there are a variety of alignment strategies, some more sensitive to a ‘richer’ understanding of values—such as Anthropic’s idea of Constitutional AI—and some that rely on more institutionalized and explicit formats such as preference satisfaction. For simplicity, I keep the dialectic in the format framed above because if it is not emblematic of the variety of alignment strategies researchers in the field use, it is nonetheless a

The structure of this paper will be as follows. In Sect. (1.1), I provide further background and motivation for the RAP and how it represents the amplification of pre-existing problems rather than the emergence of an entirely new form of power. A key culprit here is the development of behaviorist methodologies in the twentieth century and its refinement in the digital age. Section (2) introduces the alignment problem in its more standard formulation and the model of alignment on which it is based, which is one of ‘thin’ human compatibility achieved through a ‘preferentist’ model of goals and desires. Section (3) zooms out to understand the epistemic foundation from which LLMs and AI work and the modalities through which they ‘see’ the world. It will be suggested therein that reverse alignment is not (or not simply) a reflection of the strong financial incentives that commonly motivate the manipulation of human behavior but emanates from distinct constraints exemplified by contemporary ML and LLM architectures. This section explores the mechanisms through which the human side of the dynamic is modified to achieve alignment, with a specific focus on notions of value capture, such as the one presented in C.T. Nguyen’s work. Section (4) dovetails the discussion to explore the normative implications of the RAP and explores how the RAP organizes the problem space differently than other formulations—joining a chorus of researchers who collectively seek to overcome perceived deficiencies in the alignment discourse and transition to a more satisfying research avenue through which human–AI interactions can be more effectively understood.

1.1 Background to the reverse alignment problem

One thing to make clear with the RAP at the outset is that it is a species of a wider theoretical genus seeking to reduce the complexity and uncertainty of a target domain by deploying models that strip reality to its ‘essentials’. In this respect, the RAP sits in a longer history of reductionism that intertwines behaviorist theories, on the one hand, of restricting one’s analysis to observable features; and practical implications, on the other, of AI applications with their overt focus on datafication, quantification, tractability, and restricted vision

Footnote 1 (continued)

reflection of the wider economic landscape in which these technologies have been shaped, trained, and developed. Another way of saying the same thing is: having been nurtured and trained in the cradle of surveillance capitalism (Zuboff 2019), should we expect LLMs, let alone autonomous AI agents, to somehow ‘know’ these structures might be misaligned? And if so, what reason do we have to believe they will not simply reinforce some of the more pernicious ends of this economic logic? These are questions explored throughout this paper.

to directly measurable features.² To this end, two main inspirations that motivate the RAP are Shoshana Zuboff's work on surveillance capitalism (Zuboff 2019) and James C Scott's (1998) *Seeing Like a State*, which analyzes modernist state ideals and apparatuses. The two works, with their different genealogies, share a conceptual commonality, to wit, how large-scale systems 'see' their target populations in terms of proxies, metrics, and standardizations. These proxies are presumed to stand in for the most essential features of otherwise local complexities (Merry 2016). Taking metrics as exhaustively representative of groups or individuals introduces epistemic distortions, as an infinitely complex reality can be impossible to represent in standardized categories amenable to algorithmic efficiency (Deborah et al. 2021). Extending this argument, Shoshana Zuboff (2019) and Kate Crawford (2021) have independently argued that modern sciences of machine learning inherit an intellectual trend from twentieth century theories of behavior. By restricting machine learning largely to observable and modifiable behaviors, the sciences of AI risk recapitulating behaviorist theories of human decision-making and meaning-making.³

More perspicuously, behaviorism aspired to a level of technical rigor that exploited the methodologies of the time: directly measurable features and the bracketing out of inner mental states to explain animal and human behavior. This has tended to reify a notion of *homo economicus* that establishes a connection between human desires and so-called 'revealed preferences' (Amartya 1973; Anderson 2001; Franklin et al. 2022; Klingefjord et al. 2024; Millière 2025). As it happens, a form of alignment called 'preference

satisfaction' that depends on the satisfaction of revealed preferences has historically informed the alignment debate. Although alternative alignment models have been, and are currently, being explored, it is hard to ignore the fact that the vast troves of data required to train LLMs have relied on how users comport and convey themselves through engagement with preference satisfaction applications (e.g., gaming algorithms and the addictive potential behind them). Orienting alignment around satisfaction of revealed preferences runs the risk of creating a "See it, click it, stimulus–response" model of desires—as Frischmann & Selinger put it (2018: 5). This line of thinking will occupy us in the following section, but for now I want to highlight the insidious nature of behaviorist assumptions that lie behind much of the thinking in research on LLMs and AI systems.

One helpful example is provided by Kate Crawford's research into the science of affective computing.⁴ Crawford observes that "Techniques have been developed to reduce the messiness of feelings, interior states, preferences, and identifications into something quantitative, detectable, and tractable" (2021: 221). This reduction is normalized through standardized metrics and benchmarks to train LLM and other ML models (Koch et al. 2021). Moreover, AI and ML research tends to internally reference and benchmark on similar datasets and applications spaces which can further entrench a skewed understanding of human language, emotional cognition, and social interactions by translating a diverse reality (consisting of intentions, motivations, desires, will, and the myriad other psychological and folk-psychological descriptors we often invoke to explain ourselves and others' behaviors) into standardized categories: translations that, whatever their veridicality, have been met with massive commercial success. Theory reduction, it turns out, is not only epistemically appealing but bound to be commercially compelling as well.

This all aligns with Chris Anderson, then-editor of *Wired*, when he heralded this novel epistemic-cum-commercial moment early on. He remarked that the coming age could be seen as the 'end of theory': "Out with every theory of human behavior from linguistics to sociology", he writes, "Forget taxonomy, ontology, and psychology. Who knows why people do what they do? The point is they do it, and we can track and measure it with unprecedented fidelity... the numbers speak for themselves" (2008). That the high velocity, speed, and acceleration at which AI and LLMs move should be seen as the 'end of theory' is indeed curious if one has a historical sense for these things. On further reflection, in recapitulating behaviorism and its methodologies AI represents anything *but* the end of theory: it is the culmination

² Jonathan Penn makes this point exceptionally well: "AI has [always] been a science of industry since its beginnings" and that the entanglements between industry and AI advancements "lionized conformity by treating rule following, be it for profits or efficiency, as tantamount to thought itself", a conflation that "obscured the complexity of neural behavior, disingenuously equating closed systems with open ones" Penn (2020).

³ While both Zuboff and Scott address the reduction of a target domain via reliable, quantifiable, measurable proxies, they arrive at their conclusions from different paths. Scott was primarily concerned with top-down, high modernist state ideals that involved a tradeoff between faithfulness to local complexities and imperatives to scale. AI companies, contrastively, have a more bottom-up way of modulating human behavior without the need of imposing a specific norm or explicit guidance on users. An intriguing disjuncture between the two accounts, though I can only suggest it in passing herein, is the role *ideals* or *ideology* plays in modulating a given population or in scaling a system at the cost of local complexities. Indeed, one of the distinctive features of twentieth century modernism—and its problematic streak of failures—was the ideality with which it treated society, urban planning, population management, and so on. There is an overlap of this ideality with the epistemic ideality Zuboff and Crawford analyze, but exploring it further goes beyond the scope of this paper (but see Sects. 3–4 below). I am indebted to an anonymous reviewer for encouraging me to highlight this disjuncture.

⁴ The ability of computers to predict, recognize, detect, and interpret emotional states or valence.

of a well-documented and highly contested theory in the brain and behavioral sciences—a theory that, until the rise of machine learning, seemed to be on the way out in academic, neuroscientific, and behavioral research.⁵ More generally, theory and practice seem to have parted ways in terms of the depth each provides: behaviorist genealogies to one side, the theories advanced in contemporary AI systems are relatively slender and thin. This is an important point to note because the sophistication of these technologies might be seen as indicating a level of corresponding cognitive innovation that simply does not reflect AI architectures. Practically, speaking, whether the theory provides insight (or not) into the nature of intellection, it is the case that these slender theories have been massively successful in doing what they do: predicting, modifying, and encouraging user behavior. This divorce between theory and practice is unsettling, not least because most researchers at the highest level of AI development concede that building these systems is ‘more art than science’ (see, e.g., Yudkowsky & Soares (2025: 37); Amodei (2026)).⁶ Moreover, even if it is a thin and reductive theory, it is, *pace* Anderson, nonetheless a theory—and as Zuboff writes, even if “[t]heir theories are thin... The opposite is true of their power, which is monumental and largely unimpeded” (2019: 406). Strictly speaking, then the RAP is not concerned with a problem of control—the classic framing of the alignment debate—but problems of power: the power that resides in who must adapt and who gets to modify the world to fit their models (Foster 2023). From the perspective of the RAP, the problem is apparent: you can make yourself smarter or you can make the world stupider.⁷

⁵ The most foundational takedown of behaviorism came from Noam Chomsky who, in a review of B.F. Skinner’s work, had this to say: “What is so surprising is the particular limitations he [B.F. Skinner] has imposed on the way in which the observables of behavior are to be studied, and, above all, the particularly simple nature of the *function* which, he claims, describes the causation of behavior. One would naturally expect that prediction of the behavior of a complex organism... would require, in addition to information about external stimulation, knowledge of the internal structure of the organism, the ways in which it processes input information and organizes its own behavior” (1959: 49). Chomsky is widely considered to have sparked the so-called ‘cognitive revolution’: a scientifically rigorous but decidedly non-psychoanalytic school of thought that sought to understand inner mental states as mediating the behavior of the organism. In contrast, Skinner believed that a true science of behavior should do away with unobservables like internal states and confine oneself to the observable—and controllable—aspects of the situation: namely the environment. For a fascinating history of behaviorism and mind control techniques, see the recent study by Rebecca Lemov (2025).

⁶ As Eric Horvitz puts it, “Right now, what we are doing is not a science but a kind of alchemy” (quoted in Mitchell 2020). It should go without saying that entrusting our future to ‘alchemists’ of highly advanced—potentially autonomous—AI systems is a questionable decision.

⁷ I am greatly indebted to David Krakauer at the Sante Fe Institute for this point.

This paper contributes to the growing chorus on socio-technical alignment by exploring *kinds* of alignment beyond those based on preference-satisfaction (Zhi-Xuan et al. 2024) and normative shallow commitments (Milliere 2025). A guiding thread behind the reasoning in this paper relates back to the philosophical note on which I concluded the introduction: that machines might never need to become as complex as people, nor to even act like people. As Hannah Arendt 1998: 3) observed,

We do not yet know whether this situation is final. But it could be that we ... will forever be unable to understand, that is, to think and speak about things which nevertheless we are able to do. In this case, it would be as though brain, which constitutes the physical, material condition of our thoughts, were unable to follow what we do, so that from now on we would indeed need artificial machines to do our thinking and speaking.

She concludes on the eerie note that.

If it should turn out to be true that knowledge ... and thought have parted company for good, then we would indeed become the helpless slaves, not so much of our machines as of our know-how, thoughtless creatures at the mercy of every gadget which is technically possible, no matter how murderous it is (ibid.).

What this might look like, Arendt explores in her book *The Human Condition*, is provided at the end of her book when she concludes that “the trouble with modern theories of behaviorism is not that they are wrong, but that they could become true, that they are the best possible conceptualization of certain obvious trends in modern society” (2020: 322). The riddle at the heart of the RAP is this: what would it mean for behaviorism to ‘become true’ and, having transgressed the clumsy technological barriers of twentieth century behavioral sciences, do the modern sciences of machine learning risk fulfilling some of the darker promises of mid-century social engineering?⁸

⁸ This riddle is something of a Gordian knot for the RAP, admittedly. As an anonymous reviewer has helpfully pointed out, preference-satisfaction models of alignment *have* been wildly commercially successful. If behaviorism is false, it becomes harder to explain why it seems to have been so successful in its predictions. Skinnerian behaviorism and twentieth century technological limitations seem to have been the chief barrier, then for behaviorism being true in Arendt’s time. It could, thus, be reasonably argued that behaviorism was as true then as it is now—with the wonders of technology simply illuminating what was previously seeped in mystery. With the RAP, I hope to encourage caution with this interpretation; however, because of various ‘looping effects’, as Ian Hacking phrases it, that come into play when institutionalized categories and modes of behavior are produced that humans internalize to then bring those categories further into existence. Kate Crawford calls attention to this when she notes that behaviorist methodologies geared into what AI could do and amplified the theory, establishing and enhancing its hold over

In the following section, I zoom out to explore the problem space as organized through pre-existing formulations of value alignment. Emphasis will be placed on the reliance of revealed preferences and preference satisfaction as a dominant orientation through which alignment could be achieved. However, I will also look at other forms of alignment such as recent approaches on so-called ‘Constitutional AI’ and reinforcement learning through human feedback (RLHF).

2 Value alignment and thick human compatibility

The question of how to align machines with the operations of humans is not new and whether the explicit rule following required of algorithms constitutes a model of the human mind is as old as modern philosophy (one is reminded of Hobbe’s equation of reasoning with a ‘reckoning with consequences’). What is new is the unprecedented ability for machines to stand in not just for the operations of the body, then but for the operations of the mind. The reason this presents a particularly striking modern dilemma is because the idea that *human* intelligence can be separated from value-laden judgment is not particularly tenable. For the first time, though, it appears we have a sandbox case in which this ‘reckoning with consequences’ can, somehow, be dissociated from value in a way “no longer guided by human purposes” (Winner 1988). That is, a situation in which intelligence parts company from judgment and its value-laden decisions.⁹ The consequences of misaligned intelligences and misaligned values are explored predominantly under the heading of the value alignment problem, which is the social and engineering problem of specifying the nature of the values to be pursued—often called the *specification problem*—and ensuring the designed agent reliably follows steps (means) to pursue goals (ends)—often divided into the *inner* and *outer alignment problems*. This section schematically overviews alignment techniques deployed in the field of AI. While particular attention is given to preference-satisfaction models of human desire, I will also assess other approaches

like RLHF and ‘constitutional’ AI through the lens of human compatibility (Russell 2019).

To begin, the specification problem often forms the central conceptual axis for alignment. This problem emerges from the difficulty (sometimes impossibility) of precisely specifying, in an algorithmically amenable manner, the nature of the goal to be pursued. Goal divergence can lead to what in the literature is called ‘specification gaming’ (Krakovna 2018). Specification gaming is a form of reward hacking, whereby the algorithm “behaves in a competent yet undesirable way which gets high reward according to the original function” (Ngo et al. 2025). Seemingly innocuous misspecifications can lead algorithms to pursue myriad options to arrive at the goal regardless of the steps taken. For this reason, it is not simply that the goal must be well-specified but, presumably, the steps taken to achieve it should resemble those a human might take. That is, we do not want simple alignment but human compatibility (Russell 2019).

A common complaint in the literature is that specification gaming—and reward hacking more generally—stems from an inherent vulnerability of the systems themselves: they lack an *understanding*¹⁰ of what they are even doing. While this point is interesting, it is unlikely to be particularly compelling for AI researchers because no one has convincingly demonstrated what ‘understanding’ is. In fact, though, the ambiguity of the term reflects the ambiguity of the process. How humans understand the world rests on a suite of faculties and capacities that we might call intuition for convenience. The situation cannot be ameliorated through a reinforcement learning paradigm alone (Bender and Koller 2020): trial and error through iterative feedback and reinforcement learning through human feedback (RLHF) only go so far in bootstrapping more robust aspects of human understanding and intuition. Moreover, RLHF is still a form of supervised learning that, historically at least, relied on interactions between data annotators (usually underpaid, underrepresented, and actively exploited) and the users of LLMs generally (Hao 2025). The kinds of normative commitments that emerge from this form of alignment are too shallow to mitigate risks when scaled up. As Millièrè writes (2025): “Unlike humans, LLMs lack a capacity for normative reasoning to resolve conflicts rationally; instead, existing alignment methods merely reinforce shallow behavioral dispositions that can easily be exploited” (2025: 2).

More generally, it is worth making a brief aside here to relate this recalcitrant problem back to the philosophical discussion above: the structural and cognitive dissimilarities between computers and humans being significant enough to generate these misalignments—LLMs or embodied. For starters, they do not inhabit the same causal world nor

Footnote 8 (continued)

the behavioral sciences—a reading consistent with Rebecca Lemov’s work on the topic. In any case, whether behaviorism was as true then as it is now leaves the point against Chris Anderson untouched: either behaviorism has become true as a theory or machine learning has consummated what was incipiently correct all along. In either case, we have nothing like the ‘end of theory’.

⁹ For space considerations, I have to sidestep canvassing the myriad ways to interpret value beyond broad-strokes consideration of human flourishing. That said, a more in-depth analysis comes in to view in Sect. 3 when I talk about value capture.

¹⁰ John Searle (*op. cit.*) was arguably the first to popularize this view.

interact with it through the same sensorimotor contingencies that humans do. Their world does not come structured the way it does for us. They do not have built-in intuitions—folk physics, folk psychology—that allow them to fluidly engage their environment. They do not have sensors or effectors like we do. All this must be built in, programmed, and learned. If I specify to my friend not to let intruders in the house while I am away, I likely do not need to specify what constitutes an intruder. He would understand. My dog can tell an intruder from a visitor. He must learn this, but this learning does not come through obviously explicit commands, orders, or specifications. There are commonalities in shared theories of mind and shared realities that establish baselines for understanding others. There is a lot that can be communicated through ambiguity, if by ambiguity we mean non-explicit or inarticulate features of meaning-making: the parts that cannot (or not so easily) be formalized in a manner amenable to computation. None of this—one cannot be more explicit on this point—is true for AI generally speaking.

In fact, this concern converges on a foundational problem in computer science: the frame problem.¹¹ The frame problem is the difficulty and challenge of determining what information is *relevant* in any given action. What is relevant to human perception is partially determined by the fact that we are fleshy, vulnerable, precariously embodied, finite beings: we must act in specific ways to be the kinds of being we are (that is, living ones). None of that is of relevance to the disembodied and boundless intelligences of large-scale AI. As John Haugeland put it, the problem with AI is that it does not ‘give a damn’. AI not ‘giving a damn’ or, more euphemistically, *caring* arguably establishes the frame problem (Doctor et al. 2022; Harrison et al. 2022). The frame problem arguably prefigures the alignment problem. Both concern whether the fluid, fluctuating, unspecified, unfixed, and qualitative dimensions of perception and value can be translated into their opposites: fixed, algorithmically determinable, explicit, quantitative commands on which AI can be trained and to which they can be aligned. These considerations are all relevant to the RAP if we frame alignment through the aperture of human compatibility.

The alignment literature is overwhelmingly concerned with catastrophic risk scenarios in which there is a dramatic divergence between modeling human values and desirable outcomes.¹² That the designers themselves might

not properly know the thickness of the objective only compounds the difficulty of the problem. Consequently, these systems must navigate a world of redoubtable complexity and uncertainty that increases the stakes of introducing misaligned technology. Expanding the discussion further, Hubinger and colleagues (2021: 1) pivot towards understanding kinds of alignment by distinguishing *inner* and *outer* alignment problems. The outer alignment problem pertains to the ‘objective gap’ between the goal of the machine and the intended goal of the designer. This is what commonly comes to mind when one thinks of the alignment problem. The inner alignment problem is more technical in nature, pertaining to the optimization structure within the machine learning algorithm itself, that is, aligning the many subgoals a system might need as internally coherent to achieve a global (or outer) goal. In their words, “The terminology is motivated by the fact that the inner alignment problem is an alignment problem entirely internal to the machine learning system, whereas the outer alignment problem is an alignment problem between the system and the humans outside of it (specifically between the base objective [of the algorithm] and the programmers intentions)” (2021: 1). While the technical details do not concern us herein (see Hubinger et al. (2019) and Ngo et al. (2025) for relevant reviews), their framing is important from the perspective of the RAP as both inner and outer problems are treated as engineering problems—a view which is not so much incorrect as it is incomplete. To anticipate the discussion below, satisfying inner and outer alignment does not solve the alignment problem, writ large. Edelman et al. make this point particularly well: “In each case, the AI systems are locally aligned with the operator’s intention but misaligned with the interests of broader society” (2025: 2). The RAP expresses this idea further. It is a kind of meta-alignment or outer-outer alignment problem, as it were, that interrogates whether alignment between the ‘base objective’ and the programmer’s intentions translates to social alignment and the promotion of human potential and human flourishing.

The immediate implication to draw from this is that human compatibility can be thick or thin (Foster 2023). As it is commonly formulated (e.g., in Russell (2019)), it is rather thin. The reason for this is due to the model of human values and preference satisfaction on which alignment discussion is predominantly based—informed as it is by the positive social sciences (game theory and economics).¹³ This

¹¹ It is an odd fact that the frame problem, for all its historical relevance, is scarcely referenced in the alignment literature, as it arguably prefigures much of the debate—even if they are sensitive to distinct concerns.

¹² What are ‘actually desirable’ outcomes? What is desirable for the stewards of these technologies is not inherently desirable for societies in which they are embedded and from which they extract their resources. There are incremental steps on the way—gradual disempowerments, as Kulveit et al. put it (2025)—that pave the way for

Footnote 12 (continued)

misaligned AI. These gradual disempowerments are often overlooked in the alignment literature until only recently (e.g., Edelman et al. 2025). For now, I bracket this concern and return to it in a later section.

¹³ For reasons of space constraints, I have to refrain from introducing another layer of history revealed in the magisterial work of Paul A. Erikson and his history of game theory: *The World the Game Theo-*

is often called revealed preferences or preference satisfaction as it bases rational decision-making on observable manifestations of (assumed but bracketed out) underlying desires. Stated differently, revealed preferences are commonly seen as veridical enough representations of an agent's desires.¹⁴

This presents two interrelated problems: first, it overemphasizes the satisfaction of preferences as the *modus operandi* of the AI system—which is precisely what is most vulnerable to reward hacking. The second, related, problem then is whether and to what extent revealed preferences accurately reflect the inner states or desires of the agent. Is there instead a more reliable proxy of inner states on which AI inferences can be made? From the perspective of the RAP, treating preferences as the right level of granularity expands the opportunities for people to be manipulated and manipulable through reward and utility hacking. This is precisely what we know is achieved through recommendation algorithms. Preferences, then, turn out to be an ideal mechanism through which reverse alignment is effectuated, which will concern us in the coming section.

Zhi-Xuan et al. (2025) and Edelman et al. (2025) articulate the limitations of these 'preferentist' models of human behavior and alignment satisfaction. For one, preferences bundle values with other signals indiscriminately (Edelman et al. 2025: 4): "Preference orderings can carry information about anything—impulse purchases social pressure, addiction, values, momentary fads—and in their most common formulation, when they gather revealed preferences, they do in fact bundle together everything that finds its way into observed behavior—without anyway to differentiate". By way of example, if someone prioritizes a career over their affective relationships, the preferentist model is completely unable to distinguish whether this preference stems from internal ambition or external social pressure (ibid.). The temptation to understand human desires through the lens of observable, revealed behavior is tempting largely because it is epistemically 'crisper' than the 'squishier' notions of human 'well-being' and 'flourishing', let alone the subjective deliberations that inform the endorsing or coercion into any one preference choice (Nguyen 2024).

This is what makes current attempts at human compatibility thin: it places the detection and satisfaction of an individual's revealed preferences as the primary and universal

goal of the AI system (Foster 2023: 418). Instead, we should be exploring an understanding of alignment and human compatibility that, as Foster puts it, embraces a "thick and demanding world of human capacity, social complexity, and local politics of the ... pliable, universalizing world of individual preferences" (ibid.). Thickening our understanding of human compatibility has been furthered in the work of Edelman et al. They propose what they call *full stack alignment*, a pluralistic framework that does not impose any one vision or singular vision of human flourishing, "but rather seeks to prevent sociotechnical systems from collapsing the diversity of human values into oversimplified metrics" (2025: 2). The RAP is exclusively concerned with this collapse. It is concerned with the allure and seduction of metrics, standardizations, quantifications, simplifications, and the corresponding manipulation such strategies invite to make societies, people, the world fit the more predictable mold and determinate imaginaries of AI technology companies. We are the 'environment' of these AI systems, and correspondingly the chaotic and 'wilded' element of human behavior that defies immediate representation is its own kind of problem—not to us, of course, but to the creators of this technology, and, indeed, to the technology itself: how AI 'sees' the world. As Edelman et al. note (ibid.), we need a way to represent values, norms, and their interrelationships that are legible to both AI systems and the governmental, societal institutions in which they are embedded.

Before continuing, it is worth taking stock of what we have just discussed. Alignment has historically been oriented around a preferentist model of desire satisfaction. This model is not only theoretically dubious, but ethically questionable when it comes to aligning machines to human values—mostly because it is unclear whether revealed preferences are the right level of granularity for understanding the nuances of human values. RLHF for LLMs goes some way in addressing this situation because it relies not just on revealed preferences, but on explicit preferences—probing and encouraging us to train and nudge these systems towards the 'correct' answer. Whether this side-steps issues of value capture, though, is up for debate for two reasons. First, we can look at the preferentist model as the *modus operandi* prior to the development of LLMs and autonomous AI agents. This model has already shaped and warped human preferences, as we will see below. In other words, having accommodated ourselves to a preferentist model, it is hard to discern whether our engagement with more sophisticated models represents a starkly different approach. After all, sycophantic LLMs and AI-triggered psychosis seem, in part, to result from exploiting similar vulnerabilities of human psychology that gaming algorithms relied on in the social media era. Second, LLMs likely use a mixture of both revealed preferences and explicit preferences—and it is the

Footnote 13 (continued)

rists Made (2015). We have largely inherited a world that, as the title suggests, the game theorists made. Discussions of alignment do not proceed within a vacuum but within that created world. The question of course is whether the understanding of that world made through game theory is the best or more comprehensive one to conceptualize human behavior and social activity.

¹⁴ I hope the connection to behaviorism is clear enough here, but if not, it will become clearer as we proceed into Sects. (3) and (4).

need for explicit training that we know can lead to value capture, as we will see in the following section.

One novel approach to ethical AI, however, might seem to defy both RLHF and preferentialist models of value and that is the much talked about ‘Constitutional AI’, most notably associated with Anthropic’s Claude. As the Anthropic white paper describes it: “As AI systems become more capable, we would like to enlist their help to supervise other AIs. We experiment with methods for training a harmless AI assistant through self-improvement, without any human labels identifying harmful outputs. The only human oversight is provided through a list of rules or principles, and so we refer to the method as ‘Constitutional AI’” (2022).¹⁵ Constitutional AI certainly sounds like it has a lot going for it. In an almost Aristotelian manner reminiscent of virtue ethics—historically, not the most popular approach in philosophy of AI, that pride of place being given to deontology and utilitarianism—Constitutional AI is oriented around a description of virtues for Anthropic’s Claude to emulate. This is not the place to delve into the moral philosophy of this new-fangled approach, but it certainly merits some remarks as it does, on the surface, seem to embody a richer understanding of value—or virtue, even—than the approaches overviewed in this section. It will also help frame the RAP in what is to come.

Constitutional AI, as the name suggests, finds its conceptual origins in the idea of democratic AI—the idea that everyone should have an input in how these systems are trained and deployed. It consists of an initial supervised phase involving self-critique, revision, and fine-tuning before proceeding to what Anthropic calls reinforcement learning through AI feedback (RLAIF), that is, they “train a preference model from this dataset of AI preferences” (Anthropic 2022). The key benefits are numerous, but one of them is scalability—not only does it require fewer human labels than traditional RLHF, but it makes complex models easier to train. It also has the benefit of encoded principles etched into its constitution, chimerically but somehow coherently consisting of a variety of charters like the UN Declaration of Human Rights, among others.

Whether Constitutional AI amounts to a more human compatible form of AI alignment is still an open question. However, it does unfold into a concern of the RAP: what, precisely, are we aligning? The RAP has two central concerns: first, it is the way in which the human side of alignment is being modified to fit machine architectures, which is the focus of the remainder of the paper; but second, and relatedly, it also shifts attention away from autonomous technology or superintelligent AI towards (e.g., Bostrom 2014)

loci of power and agency that operate prior to, during, and after the development of this technology. Framed in this way, Constitutional AI, for all its promises, might leave the second concern essentially untouched. If the RAP is concerned with misaligned power structures—governments, institutions, companies—then one way to think about Claude’s Constitution, writes tech commentator Jill Lepore, “is that it is what happens when the state collapses. It’s because the U.S. constitution has failed that Claude has a constitution, which is apparently all that stands between American citizens (and foreign nations) and the overwhelming force of the United States military” (2026). Constitutional AI, which “is inseparable from the political events, and the constitutional unravelling, of the past decade” (ibid.), might just be a new form of the poorly executed self-regulation that has long governed the Valley’s attitude towards oversight (Bradford 2023). If anything, Constitutional AI represents a kind of swan song of self-regulation. While Anthropic represents a singularly unique exception in a vicious field of surveillance capitalism, it remains to be seen whether the company can reconcile its desire to develop responsible AI with the uncompromising demands of surveillance capitalism that other companies—not so obviously committed to the project of responsible AI—will exploit as needed to get ahead. For now, I leave these alignment approaches to turn next to the social construction of data and the epistemic foundations through which AI ‘sees’ the world and to which we are potentially being aligned.

3 Seeing like an AI

This section provides a schematic overview of how AI ‘sees’ the world. It is a world of token representations, statistical aggregates, universalized benchmarking, quantified metrics, standardized substitutions. These do not inherently limit AI. Quite the opposite, as we know: The proof is in the pudding after all, and the performance of these systems, we could say, ‘speak for themselves’. This section unpacks these claims further and identifies how behavior and qualitative aspects of the human condition are rendered into machine-readable and machine-legible forms. This section proceeds in two steps: first it looks at the reliance on benchmarking and metrics as stand-ins for the variegated human reality they aim to represent; and second, proceeds to identify value capture (Nguyen 2024) as the main mechanism through which institutions represent their target domain. The insinuation to be made more explicit in Sect. (4), then is that AI inherits this kind of institutionalized gaze—such that we can borrow from James C. Scott’s analysis of large-scale modernist analysis of ‘seeing like a state’ and translate it into ‘seeing like an AI’.

¹⁵ The paper is linked to their website, here: <https://www.anthropic.com/research/constitutional-ai-harmlessness-from-ai-feedback>.

3.1 How AI ‘sees’ the world

Algorithms institute particular worldviews through data. Data is not simply found; it is not lying around as a kind of ‘natural resource’. It is created and rendered. It comes through rigorous categorization processes and the unique modalities and infrastructures of digital technologies. When we think of datasets, we tend to think of digital representations of a neutral field of objects. But those datasets had to be created, compiled, and categorized. For a long time, this relied on (usually underpaid, underrepresented) data annotators: the laborers behind the large-scale technology apparatuses that produce new products at a breakneck pace. This means data is not found but rendered from human experience that is viewed, from an economic and technological lens, as a ‘raw material’ in need of harvesting and channeling to specific ends. As Boyd writes, “What data exist—and what do not—are shaped by social choices” (2023: 237).

Given that social choices permeate what data structures exist, it is important to assess the source and quality of data used in machine learning methods, which are dependent on institutionalized standards. To an extent, this will always be inevitable since institutions need a high level of granularity to ensure consistency. Nonetheless, while these are not flawed methods in themselves, they are more socially constructed than terms like machine learning and artificial intelligence suggest, which tend to invoke a kind of ‘view from nowhere’ or ‘God’s eye view’. The aspirations towards such a perspective belie the fact that “technology is consistently leveraged to codify the values that its maker or users wish to make rigid” (Boyd 2024: 328). Machine learning systems do not (or not simply) depend on these metrics and institutionalized gazes due to our fascination and trust in quantification (Porter 2020). Rather, it stems from the very nature of the system in question. They require structured data—proxies of the world—to get off the ground.

If one replies that we rely on structured data, too, that simply misses the point. All data are structured. The ‘data’ that informs our perception, if data is even the right language to speak of here,¹⁶ is structured because our sensory apparatuses, the brain, and the materiality of embodiment itself are organized in such a way and such a way as to channel flows of energy and sensory information—somehow—into (un)conscious representations. That is different from pre-structured and prepackaged data that was compiled by human data annotators whose primary animus is to render human

behavior predictable, regular, and less spontaneous. These ‘predictive imperatives’, as Zuboff calls them, establish the networks through which surveillance capitalism operates. It is surveillance capitalism that has served as the cradle out of which modern machine architectures, corporate logics, and economic imperatives emerged. I am interested in how the institutionalized gaze that has served as the cradle for modern LLM systems is recapitulated in these novel technologies—whether LLM ontogeny recapitulates a kind of surveillance capitalist phylogeny.¹⁷

Instrumental to the thinking behind the RAP is the work of Kate Crawford (2021), who, as a senior researcher at Microsoft and experienced data scientist herself, has a unique perspective on not only how corporations like Microsoft and Amazon work, but also how machine learning systems operate. Machine vision, for instance, operates over images without much context, often ignoring the natural or social situations that would inform how humans perceive images. Contextual background conditions are harder to formalize and specify. It is far simpler to identify a proxy for those background conditions and train a model on that. This is apparent in how machine vision systems operate in the contentious cases of facial recognition and predictive policing (Moore et al. 2025; Narayanan & Kapoor 2025). The earliest such systems were trained on a national database of mugshots, reducing the persons behind the crime into a series of decontextualized images in the name of solidifying a kind of ‘physiognomic logic’—a logic that sought to interpret criminal dispositions based on overt physiological or behavioral features and facial expressions (Narayanan & Kapoor 2025; Bender & Hanna 2025). These images—persons reduced to patterns—contain people with “rich personal histories, structural inequities, and all the injustices that have accompanied legacies of policing and prison systems” (Crawford 2022: 94). More generally, Crawford notes how “[A] computer vision system can detect a face or a building but not why a person was inside a police station or any of the social and historical context surrounding that moment. Ultimately, the specific instances of data ... aren’t considered to matter for training an AI model. All that matters is a sufficiently varied aggregate” (ibid.). Context is not inherently germane to the training of these systems. The point that contemporary systems—machine vision, natural language processing or otherwise—rely heavily on a ‘sufficiently variegated aggregate’ cannot be overstated as it expresses the belief, ubiquitous in computer science and artificial intelligence research, that such an aggregate

¹⁶ My suspicion is that it is not the right language. Data implies a translation process. The very nature of ‘data’ is different from sensory expression. Data is a term of art that only metaphorically applies to human perception. See Seth’s (2025) recent defense of biological naturalism to which the view in this paper is sympathetic.

¹⁷ Of course, not all artificial intelligence are built equally. LLMs might not necessarily ‘flatten’ values to the same extent that a gaming algorithm might. Nevertheless, it is difficult to know in practice where the training data comes from, or how much of it comes from gamified input from user feedback.

of collated digital objects, images, and text can somehow arrive at a semantically streamlined core of criminality and identity.

This stripping of an image or a text of its context to produce ‘value neutral’ data is a social and political act with implications for the epistemic status of machine learning systems. The problem, already incipient in smaller scale systems of the early predictive technology years, has only become amplified during the AI revolution of this decade. This is because the difficulty of understanding what is in the training data and where it comes from scales with model size (Bender et al. 2021). Because LLMs scrape data from most of the internet indiscriminately using systems like Common Crawl, their training data is an internet-sized model with precisely zero sensitivity to the value-laden, normative, and contextual underpinnings that guide, inform, and mediate social or online interactions. This is also why all LLMs must undergo intensive and involved post-training processes to ensure that the right, or at least the minimal, normative guardrails are placed on them to prevent them from extruding material found in the darkest and most toxic recesses of the internet (see Millièrè (2025) and Millièrè & Buckner (2024) for insightful overviews of this process). The whole concept of ‘data mining’ is designed to abstract meaning from the patterns to arrive at variegated enough statistical aggregates to increase model functionality: burying the messages in those patterns under ever-increasing digital profundity.

This makes LLMs statistical prediction engines that do not inherently have an eye towards the meaning of the tokens they anticipate. Understanding, under this paradigm, emerges spontaneously through a voracious process with as inclusive a dataset as possible. This has implications for how machines come to interpret human behavior that, in turn, has implications for what machines are being aligned to (and, as we will see below, whether it is instead ourselves aligning with them). Returning to Russell’s (2019) proposal above, he suggests that human compatibility and alignment could be achieved not through programming direct normative constraints (beyond very general rules), but through inverse reinforcement learning: that AI systems learn our preferences through observed behavior and revealed preferences. This path is not tenable, however, for reasons Jacob Foster notes: “While AI designed to satisfy human preferences might seem as if it is merely learning our preferences, it is in fact constructing both individuals and social worlds in a thin, universalizing, atomizing fashion” (2023: 425). He rightly asks, “Where is the social, where the political, in the vast enterprise of preference satisfaction?” (ibid.).

As stated, revealed preferences rest on too thin a notion of human compatibility to prevent manipulation of human preferences that could lead to reverse alignment—manipulations we already know occur within corporate logics and

which could simply be recapitulated in, or instrumentalized through, LLMs. This all gestures in a more general direction towards a systematic privileging of quantified, standardized, and efficiently processed rendering of human behavior into predictive proxies that enable more accurate anticipations of human choices. It also systematically marginalizes or neglects (if it acknowledges at all) aspects of human experience that resist immediate or easy quantification: embodied knowledge, affective interpretations or registration of an experience, phenomenology, intersubjectivity, and social heuristics (see Birhane (2021)).

This problem is actually more acute than it initially appears. Contemporary ML practice is dominated by ‘benchmarking’. Koch and colleagues (2021) have written on the systemic nature of internal benchmark referencing. In this process, things that we want a computer to do—recognize an individual or an object in a scene—are represented in specific benchmark datasets and well-defined, if narrow, metrics for evaluating the success of a task (Koch et al. 2021). Again, pulling from Foster, we find the problem is particularly pronounced and that the costs can be substantial “insofar as benchmarks (datasets, metrics, and task operationalization) depart from the details of the real-world problem, purported progress can be illusory, and real-world deployments may go catastrophically wrong” (2023: 419).

This industry-standard process presents several intriguing philosophical issues: task communities create their own datasets, and these datasets are borrowed from other tasks, placing into question the ecological validity of these algorithms when translated across domains. Adding to this worry is the fact that, according to Koch et al.’s survey (2021), only 12 top universities and companies produced the benchmarks used in twenty-six thousand papers. Here it bears repeating Boyd’s remarks above, namely that this technology is consistently used to codify the values and trends that its makers want to make rigid (2023: 328). If the mapping out of fields of entire objects—the purported goal of compiling these datasets in the first place—is relegated to a handful of highly powerful, highly centralized, highly geographically specific (largely anglophonic) locations, then it is in fact pertinent to ask what worldviews are packaged with those datasets. This is what makes current trends in AI alignment human incompatible: they rest on a “handful of massive unrepresentative datasets and gigantic ‘foundation models’—standard, centrally produced solutions to core algorithmic challenges that serve as building blocks for larger systems. This mode of AI practice demands standardization and legibility; it is only through such strategies that engineers can reach the scale demanded by current architectures” (Foster 2023: 420).

What this suggests—and here we circle back to the RAP—is that machine learning relies extensively on behaviorist methodologies that recall twentieth-century theories of animal and human cognition. The most emblematic

researcher of this line of thinking, of course, is B.F. Skinner—though his student Paul Ekman did much in the way of extending behaviorism to the domain of affect and emotion recognition. Skinner prioritized behavioral features that are directly observable to third-party viewers and remained neutral on the inner workings of the mind in explanations of behavior. What was relevant to determine the cause or elicitation of a certain behavior was the animal's dispositions as related to its learning environment (usually to identify a set of hard-coded, stereotyped responses or learning rules). Historically, of course, this contrasted with then-popular theories of mind like Freudian psychoanalysis that were increasingly seen as pseudoscientific and, as such, placed psychology in a less-than-positive light. By restricting the lens of study to directly measurable, behavioral, and predictable features, behaviorism aspired to place the study of animal behavior on the same scientific level as the 'hard' sciences—physics being the ideal target because of its elucidation of universal laws and the predictive power those laws generate.

What is surprising about the story of behaviorism is how it was actually on the way out in general psychological research in the twentieth century. Its persistence in the study of human emotion remained entrenched beyond behaviorism's decline, due to Ekman's studies on human emotion and its universal, culture-independent expression, but it, too, was also exiting stage left in the study of affect and emotion in cognitive neuroscience. It is an intellectual curiosity that it has been resuscitated and reanimated with such fervor in the sciences of machine learning and machine behavior. An intellectual curiosity, but not surprising: the reductive tendencies of behaviorist thinking found fertile ground in the emerging digital technologies of the early twenty-first century. With a desire to map and anticipate a world of objects—where the differences between the subjective and the objective, the animate and the inanimate, were collapsed—a crude stimulus–response model of human behavior suited the (comparatively) simpler architectures of these initial predictive systems and the machine readability that the stewards of this technology relied on to turn their products from simple services—like Google Search—to genuine financial bulwarks.

Writing about the history of the Internet of Things [IOT], Zuboff makes the connection with behaviorism explicit: "Ultimately, IoT's true value depends on customers adjusting their behaviors and risk profiles based on feedback from their 'things'" (2019: 215). These systems are by necessity "trained on measurable action, it only cares that whatever we do is accessible to its ever-evolving operations of rendition, calculation, modification, monetization, and control" (ibid.). In behaviorism, a theoretical narrowing to directly observable features enhanced its 'scientific objectivity' particularly in enabling, or so it was hoped, the predictability of the

animal. In machine learning, these predictive imperatives are internalized to enable predictive 'nudges' to encourage users to behave in predictable, if also tailored and bespoke, ways. That this is an ethical worry for social media technologies and recommendation algorithms is particularly pronounced, but its philosophical and epistemological consequences often go less appreciated. Campolo and Crawford make the epistemic consequences of this 'flattening' pronounced: "[This] epistemological flattening of complexity into clean signals for the purpose of prediction is now a central logic of machine learning" Campolo and Crawford (2020).

Understanding these maneuvers as a kind of epistemological flattening brings into focus the perspective—such as it is—that AI systems can have. An example from affective computing will help clarify the argument. Affective computing is the ability to detect, interpret, and predict emotional expression and infer an underlying mental state or behavioral disposition accordingly. The theory that motivates affective computing is that of Paul Ekman's theory of emotions. Ekman is known for his work on the 'basic emotions': happiness, sadness, fear, disgust, anger, and enjoyment. These emotions were posited not only as universal but also to have corresponding facial and physiognomic expressions across cultures. Iterations or variations of these emotions might be more or less finely tuned depending on cultural context, circumstances, and contingencies—but these specificities were seen as emanating from a core, context-neutral emotional reality. As he would write, "particular facial behaviors are universally associated with particular emotions" (Ekman and Friesen 1978: 128).

While affective computing might *prima facie* appear different from behaviorism because it references specific 'inner' emotional states, it is figured in the behaviorist paradigm because it draws a direct linkage between emotional states and facial (behavioral) expression. The thickness or thinness, the meaning and the semantics, of an emotional state—to say nothing of the enveloping mood that might accompany it—are bracketed out. What mattered for Ekman, and what matters for affective computing, is the physiological indication of an intention. But all this rests on a behaviorist assumption that what is observed is a representation of the inner subjectivity of the one being perceived. As a famed neuroscientist of emotion, Lisa Feldman Barrett writes, AI systems might be "trained to detect a scowl [but] that's not the same thing as detecting anger" (Barrett et al. 2019).

It might be argued that the case of affective computing departs from the discussion of preference satisfaction around which much of the alignment discourse turns. They are certainly attendant to different aspects of the predictability of human behavior, but they are based on a similar underlying assumption and thus the former provides a useful vignette for understanding the latter. Indeed, despite Ekman's theories of emotion being extensively debated and challenged in

affective neuroscience (Barret et al. 2019), “In the AI field, Ekman is commonly cited as though the issue was settled, before directly proceeding into engineering challenges. *The more complex issues of context, conditioning, relationality, and cultural factors are hard to reconcile with the current disciplinary approaches of computer science or the ambitions of the commercial tech sector*” (Crawford 2021: 169; emphasis added). To make the connection with the alignment literature—geared as it usually is towards preference alignment—the idea that human preferences are accurate or veridical representations of human goals and desires has been extensively challenged (see Anderson and Anderson 2007; Zhi-Xuan et al. 2025) and the context or inner motivation driving the pursuit or endorsing of a certain preference (and, in turn, whether it is satisfied) is also largely omitted from preferentialist accounts. Both cases gesture at a common theme regarding alignment and machine perception: namely a contested theory fit what AI tools could do: “Ekman’s theories seemed ideal for the emerging field of computer vision because they could be automated at scale” (ibid.: 175). This need for scale animates much of the discussion of reward, attention, and value hacking that we will explore in the following section. The issue of value capture, as we will see, and how it relates to value alignment through revealed preference satisfaction establish the central dynamic for the RAP: that human behavior be made to fit the tools of machine learning, machine vision, and large language models and that, in turn, we come to understand ourselves through these lenses.

To summarize, the RAP at its core captures a marriage of convenience between machine architectures, on the one hand, and the economic incentives that underpin their deployment at scale. As we have seen, LLMs and other machine learning algorithms rely heavily on benchmarks and foundational models that impose a particular semantic stamp onto the world: “Any person or organization that needs to interface with these systems benefits from aligning with their classificatory schemes, their way of seeing the world” (Foster 2023: 420). The discrepancy between the values over which we deliberate and endorse and the prepackaged classificatory schemes that might be easier to internalize and adopt creates the central tension of the RAP: rather than AI systems expanding to accommodate the full range of human experience, we are increasingly modifying and simplifying values into more machine-readable forms and becoming more algorithmically legible in the process. When we adapt ourselves as ‘data subjects’, as quantifiable self-presentations, algorithmically visible, and computationally efficient, we are changed in the process. The result is that algorithms only see these data subjects and potentially remain blind to innumerable others. The result is not merely a technological limitation but a notable reshaping of human expression and values to fit within the perceptual

constraints that computational systems can process. The next section will make the core mechanism through which this is achieved more explicit: that of value capture as proposed in the work of C.T. Nguyen.

3.2 Value capture and predictive imperatives

Directly preceding this, I described a process whereby we become data or digital subjects, attuned to the algorithmic legibility that mediates our online interactions. C.T. Nguyen helpfully introduces the concept of ‘value capture’ to help make sense of this process of translating human experience into machine readability. This section, then, delves deeper into the notion of ‘value capture’ to explore how the value landscape—especially when formulated through the lens of preference satisfaction—is at risk of being datafied, digitized, quantified and, in effect, gamified to render it more predictable, measurable, manipulable. This section works in conjunction with the previous one in that it highlights how the machine architectures that are currently developed at scale operate according to incentives laid out in surveillance capitalist infrastructures of value capture.

Value capture is when “an agent’s values are rich and subtle; they then enter a social environment that presents simplified—typically quantified—versions of those values; and those simplified articulations come to dominate our practical reasoning” (Nguyen 2024: 1). Certain systems can be intended to create value—such as social media putatively facilitating the exchange of ideas—that lend themselves, through bottom-up interactions and top-down incentives, to value capture. In this process, values are re-articulated to suit either (sometimes both) the economic imperatives of those who control the technology or tailor the format of values to be better aligned with digital infrastructures of interpretability and predictability. Value capture is a general phenomenon. It occurs outside and within digital economies. But it presents a unique issue with respect to value alignment and the RAP. If value capture depends on a degree of hyper-explicitness and metrification (with the proclivity to simplify the values being represented in the process), and if contemporary AI systems rely on such simplifications—if that is what they are—then that informs the interpretation of the values to which AI is aligned and engineered.

C.T. Nguyen describes value capture as proceeding in three interlinked steps (2024: 7). “First, an agent has values that are rich, subtle, or inchoate. At this stage, values are in the deliberative and developmental stage. Second, the agent is embedded in a wider, typically institutionalized context that presents an explicit value expression of some value or desire—typically simplified, standardized, or quantified. Lastly, the explicit value expression in its unmodified or undeliberated forms comes to dominate the agent’s practical reasoning and critical thinking processes in the relevant

domain.” It is important to note that value capture need not be a pernicious process of coercion: people may consciously endorse these values, perhaps because it is easier to take off-the-shelf values than undergo the more concerted process of value deliberation. Fully reflecting on, thinking through, and endorsing or not endorsing a given (set of) value(s) is cognitively demanding and usually time intensive. Speed and rapidity—the kinds of which we see through user engagement of social media platforms—remove friction that might otherwise cause us to pause and question whether we do or do not believe a certain viewpoint.

Institutionality is a focus of value capture because it typically requires ‘hyper-explicitness’ and standardization that is unlikely to represent complexity within human values (ibid.: 17). It is also why the term rendering is suitable to describe the process: rendering means to translate one form into another, in this case, yet-to-be-determined, undefined, under deliberation or undeliberated values into hyper-explicit, machine-readable formats—rendering the semantics of the former into mathematical representations of data, a process which tends to strip the former to the essentials of the interaction. Semantics might be along for the ride, but it is not inherent to the relationship embedded in the dataset. The simplification of values that comes with institutionality has consequences for how an agent comes to reflectively deliberate and endorse a given set of values. Nguyen captures this well when he writes:

Here’s another way to put it: value capture, even when consensual, involves a low degree of granular control over the details of the contents of one’s values. It puts you in the same relationship with your values as you have with, say, your iPhone’s End User License Agreement [EULA]. When you click to sign a EULA, you did, technically consent, and you are, technically, responsible. But you only have one binary choice: accept the whole package or not... This low granularity arises directly from the core functioning of large-scale collective values. (2024: 24)

This ‘low granularity’ is not an accidental feature of value capture, but rather a convenient consequence of its institutional logic—and the creation of what Zuboff calls ‘behavioral surplus’. Part of the logic of surveillance capitalism, according to Zuboff, is that its financial fecundity is predicated on establishing ‘behavioral futures markets’. These are markets where the unanticipated or spontaneous influences, which might otherwise engender unpredictable behavior, are vitiated in favor of designing an infrastructure that encourages predictable engagement with a system or a product. Reading Nguyen through Zuboff is helpful here as it clarifies how surveillance capitalism relies on the harvesting of value. It relies, in effect, on value capture. That the data harvested through means of surveillance capitalism is

the entire foundation on which all modern technologies of ML, AI, and LLMs operate is well-documented. As Karen Hao puts it, “Detailed digital trails of people’s thoughts and ideas on social media [are] merely ‘text’. People and vehicles in pictures [are] merely ‘objects’. Surveillance [is] merely ‘detection’” (2025: 102). Text, objects, detection: all of this informs the training of large-scale, hyper-complex LLMs. LLMs and large-scale systems of surveillance are animated by the same institutional logic: “Institutions need to render the world into a format legible to large-scale institutional information-processing procedures” (Nguyen 2024: 30). Nguyen continues: “Only certain things count. Institutional measures need to be usable across different contexts. This requires that the measures leave aside highly context-dependent forms of understanding and focus, for their inputs, on context-invariant qualities” (ibid.). The interesting question, from my point of view, is whether LLMs, trained on captured value, recapitulate the tendencies present in surveillance capitalism. My suspicion is that they do, but bringing the conversation around to a more full-throated justification for that belief is reserved for the next section.

The overlap between Nguyen’s and Zuboff’s analyses and that of Crawford’s is remarkable. It also establishes a direct connection between the social and economic processes of value capture evaluated in this section with the machine architectures that abound in society: “Many of AI’s achievements”, writes Crawford, “have depended on boiling things down to a terse set of formalisms based on proxies: identifying some features while ignoring or obscuring countless others” (2021: 221). Datasets like ImageNet were designed to “map out the entire world of objects” (Gergeshorn 2017), which is somewhat amusing when one considers the millennia-long history of ontologizing that has failed to settle debates about the status of beings, objects, and subjects. But these philosophical nuances are not of concern to this map—indeed, the very distinctions between subjects and objects are obscured in the ontological fixing involved in such a translation:

[S]urveillance capitalism’s new instrument will render the entire world’s actions and conditions as behavioral flows. Each rendered bit is liberated from its life in the social, no longer inconveniently encumbered by moral reasoning, politics, social norms, rights, values, relationships, feelings, contexts, and situations. In the flatness of this flow, data are data and behavior is behavior. The body is simply a set of coordinates in time and space where sensation and action are translated as data. All things animate and inanimate share the same existential status in this blended confection, each reborn as an objective, measurable, indexable, browsable, searchable ‘it’. (Zuboff 2019: 211)

To take stock, while value capture does not imply surveillance capitalism, surveillance capitalism necessarily implies value capture—and it is the distinctive imbrication of value capture, surveillance capitalism, and standardized information-processing architectures that presents a particularly thorny problem for value alignment. Stated differently, if the current stewards of this technology are the cradles of surveillance capitalism—and the extractive, predictive imperatives that make it possible—then it should at least be entertained that the LLMs we are seeking to align are aligning to an already rendered, already translated, already simplified understanding of human values: that AI and LLM technologies will inherit these imperatives, operate according to them, perhaps even amplify them. AI technologies, as they stand, are not autonomous from their creators—though they might imminently become so. If one might argue that these systems becoming autonomous implies they know better than their creators and would know what we ‘really’ want, that is simply to beg the question. It also overlooks the very nature of what they have learned. Nothing in the alignment problem—commonly construed—implies that satisfying revealed preferences is inadequate to satisfy alignment. I see zero reason to assume that these systems, having become autonomous, would even know that what they were doing before they were autonomous—in the hands of technology companies that might not have our best interest in mind—was not aligned with human goals, flourishing, and well-being.

4 The message hidden within the pattern: the reverse alignment problem

The reverse alignment occurring has hopefully been made clear by now—at least in broad, schematic form. It is now necessary to zoom in on more specific instances in which the RAP occurs. The RAP identifies a marriage of convenience between economic incentives that guide corporate decision-making on when and how this technology is implemented in the public—on the one hand—and computational architectures that, when working ‘at scale’, require voracious amounts of data that are represented as proxies, metrics, and likely simplifications of the value landscape—on the other. In any case, whether LLMs and AI more generally contribute to these simplifications, the fact remains that the technology of the preceding decade has morphed the value landscape generally. As is not hard to guess, it is hard to predict the longer term consequences of this technology and its acceleration by and through the ubiquity of LLMs—though we will explore some preliminary evidence that they are only exaggerating a pre-existing trend. The interesting question is whether LLMs—or another path on the road to Artificial General Intelligence (AGI)—inherit these trends;

whether they avoid the currency of value capture on which most AI systems learn; and whether they enhance human self-expression, social benefit, and well-being.

To recapitulate, then, the RAP is not dependent on all and every aspect of human experience resisting computational legibility or machine readability. It is not dependent on the claim that the value landscape—local values, personal deliberations, democratically negotiated projects, internationally mediated conflicts, whether I should flake on a friend tonight because I do not want to see their partner—with all its variegated textures and nuances will forever remain allusive to AI systems. The RAP identifies how, if not anticipated, AI might never need to. From the perspective of the RAP, it is simpler to ‘re-engineer’ humanity to conform closer with our mainstream computational practices than it is to overhaul an established paradigm in machine learning that has received enormous financial watersheds and shows little sign—or desire—to decelerate. This focalizes a key point in the debate: namely that alignment as has commonly been construed focuses on a rather thin notion of the human, the political, and the value-laden. This overlooks the myriad political, social, and material origins in which alignment is likely to occur. Observes Crawford: “To understand how AI is fundamentally political, we need to go beyond neural nets and statistical pattern recognition to instead ask what is being optimized, and for whom, and who gets to decide” (2021: 9). In other words, we need to understand the message hidden within the pattern.

This section unpacks a potted history of how humans have engaged with (mis)aligned technologies in the past. It begins by looking towards data mining through social media platforms and their technologies that have established systems of data harvesting, behavior modification, and value orchestration for their own ends. This is something of a precursor to the RAP, as the suspicion is that having learned our preferences through our interfacing with these technologies, it is highly questionable whether this ‘re-engineered’ (Frischmann & Berger 2018) version of humanity is the right slant for alignment. It then proceeds to show how humans are already adapting their behavior to the distinctive communicative modes through which LLMs interact with users—altering cultural and self-understandings in the process. What follows from that can only be speculative, as we do not know what autonomous AI systems with ‘superintelligence’ or AGI—as defined, as it were, by the mission statement of OpenAI¹⁸—will do if achieved. However, my argument is that, having developed in the nexus

¹⁸ “OpenAI’s mission is to ensure that Artificial General Intelligence (AGI)—by which we mean highly autonomous systems that outperform humans at most economically valuable work—benefits all of humanity” (OpenAI Charter: <https://openai.com/charter/>).

of surveillance capitalism's myriad modes of operation, it will recapitulate these more negative tendencies and encourage modifications and simplifications of human behavior. For space constraints, the argument here is almost entirely restricted to the identification and precise articulation of a problem rather than a solution (see Edelman et al. (2025) and Millière (2025) for more positive proposals); however, I reserve some more positive observations in the concluding section of this paper.

To begin, recommendation algorithms and social media platforms provide an illustrative example of an early precursor to the RAP. To see how, it is important to consider that "Social influences act through a mechanism and the character of their action depends upon the character of the mechanism" (Carr 2025: 9). Controlling and surveilling the character of that mechanism, Carr notes, are the *raison d'être* of large technology corporations. There is every incentive to control that mechanism as it enhances and enables further user interaction, consumption, and manipulation. While the character of social interaction has constantly been modified with emerging digital infrastructures,¹⁹ the acceleration of communication with the ubiquity of the internet made once-stable social structures and relations as malleable as water. This acceleration encouraged a novel way of interacting with others through the medium of a digital computer. It also encouraged the steady erosion of the more chaotic or 'wilded' elements of those interactions—a problem to be solved, and a bug to be rooted out. Here, again, Carr expresses the transformation when he writes that the entrepreneur's vision of a global community that works for everyone—the early and now naïve view of what the internet was—turned out to be a fantasy: "Human beings are not computers. The communities they form are not electronic networks. Society does not scale. What was missing from Zuckerberg's manifesto was any sense of people as individuals, with their own backgrounds, and beliefs, personalities and motivations, quirks and biases" (Carr 2025: 17). To reiterate, these latter inconsistencies with digital architectures, which work with cold, predictable precision, these 'quirks and biases' were just so much 'prediction error' for the pioneers of this technology. Of course, machine-like predictability was precisely the worry articulated half a century ago in Hannah Arendt's observation that "the trouble with modern theories of behaviorism is not that they are wrong but that they could become true... that they actually are the best possible conceptualization of certain obvious trends in modern society".

Zuboff makes this connection especially clear: "Skinner imagined that with the correct technology of behavior knowledge could pre-emptively eliminate anomalies, driving all behavior toward pre-established parameters that align with social norms and objectives" (2019). If this sounds like a histrionic exaggeration of the problem, we only need to look to the creators of this technology themselves. Writing on the history of Google, or Alphabet, Zuboff notes: "Instead of the typical assurances that machines can be designed more like human beings and therefore less threatening, Schmidt and Thrun [former CEOs of Google] argue that just the opposite: It is necessary for people to become more machine-like" (ibid.). Recommendation algorithms and social media were early experimental playgrounds through which 'behaviorism could become true'. But how?

Again, social influences operate through a mechanism and the character of their action depends on the character of the mechanism. We have already seen at least two key mechanisms through which the influence over social interaction is achieved: the elimination of ambiguity by rendering behavior into machine readability and how such rendering lends itself to value capture. Values have always been 'squishy', unclear, fluid, unfixed, subjective, heavily qualitative, highly qualified—in a word, ambiguous. The computational ability or inability to handle ambiguity is arguably the philosophical foundation of computer science: from the early days of rigid expert systems, with hand-coded rules, through to the deep-learning revolution and algorithms that focus on prediction error minimization. The difficulty of interpreting ambiguity stems from what computer systems are: information-processing systems—they rely predominantly on so-called Shannon information. Shannon information concerns how efficiently and unambiguously a signal could be communicated, and when ambiguity persisted, how it could be minimized. This focus on Shannon information would go on to inspire the early cyberneticists who identified prediction error mechanisms and feedback as the fulcrum of adaptive behavior. Shannon, however, explicitly acknowledged this did not inherently imply semantics: the meaning of a message and the transmission of that message are orthogonal questions—and much ink has been spilt trying to unify the two.

What is important from the present perspective is that Shannon's formulations enabled industrial objectives of efficiency, automation, and standardization to be rigorously applied at scale and to human interactions themselves. Working contemporaneously with Shannon, cultural anthropologist Margaret Mead questioned whether the optimization of communication through mathematical formalisms introduced its own kind of epistemic distortions: one that would obscure the sense of the messages and strip away its nuance: "As mechanisms for communication are made more efficient, would subtleties of meaning get lost or garbled?"

¹⁹ The telegraph is an early precursor to this general dynamic and when one reads of the excitement of the epochal and civilizational potential of this technology it is hard not to see resonances with our current techno-optimist age.

Would the signal, not the noise, turn out to be the more dangerous source of distortion?” (Carr 2025: 58).

Mead’s worries would become more pronounced in the social media age—the meaning of the messages becoming more deeply obscured by behavioral flows of endless data: “Content has collapsed”, Nicholas Carr writes, “as our adoption of the drab, generic term content to refer to all forms of expression testifies. Everything now has to fit the internet’s conventions and protocols, with their stress on the immediacy, novelty, multiplicity, interconnectedness, and above all efficiency” (2025: 63). As digitization has scaled society, communication has lost its human scale, putting into question the goal of such universalized scale.

For instance, when social media algorithms rank and organize information, they do so without regard to the meaning of the content: it is maximized for engagement. The question of interpretation and significance metamorphosizes into one-dimensional proxies (how long a cursor hovers over an image; the universal ‘like’ symbol ostensibly representing approval, but might also, as we know, just be an expression of simple acknowledgment). ‘Like’ buttons form the backbone of the then-revolutionary Facebook News Feed, which, as Carr notes, “was inspired by Mark Zuckerberg’s frustration with the inefficiency, the friction, of online communication, which to his mind stemmed from the involvement of human beings in the process of evaluating, selecting, and sharing information” (2025: 64). Computers, with their efficiency and programmability, contrasted with the slow and capricious human world, so the former provided a unique ‘solution’ to the latter: “The feed removed human deliberation and judgement from the process. It replaces personal agency with machine agency” (ibid.). A notable consequence of this machine agency is that, like we saw with the perspective of surveillance capitalism above, the semantic aspects—what the patterns mean to us as meaning-makers—are irrelevant for how the algorithm structures the news feed: everything is granted equal semantic weight: “Everything”, notes Carr, “had the same semantic context, which was no context” (2025: 64). This is a view in which semantics is along for the ride. It has no intrinsicity outside of the datasets and how users, as points and relationships, engage in structured and predictable ways within that dataset.

The introduction of these imperatives as mediators of social interactions transforms patterns of human thought and engagement (Ott 2023). There are cognitive consequences of this technology and there is growing evidence that it gets ‘under the skin’, as it were, by influencing the (under)development of certain neural pathways (Korte 2020). The neuroscientific consequences of this technology are, of course, provisional and difficult to anticipate, given the novelty of the technology and its largely unimpeded deployment for ‘digital natives’, but the cognitive consequences are becoming increasingly pronounced. Articulating

the significance of this, Brian Ott notes that, while shifts in prevailing technologies have always enabled shifting habits of thought, digital electronic media introduce more acute vulnerabilities for humans: “As humans attempt to mimic computer efficiency, they rely more heavily on instinct and affect. In short, as humans try to speed up their information-processing and decision-making capabilities, they are less careful and rational and more impulsive and affective, which—paradoxically—undermines the quality of decision making” (2023: 10).

These cognitive consequences (which we will return to momentarily) have an impact on our interpretation of value- and context-sensitive features of the social world. One way to see this is through the lens of moral de-skilling (Green 2019; Vallor 2015). Moral de-skilling is the process by which humans outsource conscientious decision-making and moral deliberation to machine learning systems (for instance, asking an LLM to produce a letter of condolence, apology, or self-defense). Habitual reliance on this results in underdeveloped moral reasoning skills in real-world situations when access to an LLM might be unavailable. The advent of LLMs presents a particularly striking form of de-skilling as moments of reflective and critical moral, social, and personal deliberation become outsourced to (not necessarily context sensitive, hallucination prone, and sycophantic) LLMs. As Green notes, this outsourcing may inculcate, even normalize, the de-skilling of the uniquely human ability to engage in moral deliberation and engage in democratic processes, which are context sensitive and value laden, and which might become a rarity. To put this concern into focus: the worry is not (or not simply) that we will be unable to deliberate effectively among ourselves: our ability to interpret the accuracy or acceptability of LLM or AI-generated decisions or recommendations will be impacted in turn: “very few people may be able to perceive if such AIs are making mistakes and we will be living in a world that we have made, and yet at the mercy of things we no longer understand or control” (Green 2019). These scholars are all concerned with the supplanting of human agency with a kind of machine agency, and this concern is accentuated in cases where reverse alignment occurs. Adding to Green’s concern, it is not just that we will be unable to interpret machine judgment, but that we would align with machine judgment in an uncritical and indeliberate way.²⁰

To unpack the cognitive consequences of this technology further—sticking to the case of social media, for now—media “creates the conditions that in turn condition us. The

²⁰ The pervasive view that machine logic is more rational would only add to a collective belief that delegated and outsourced systems know ‘better’ than us. I finish this paper in the following section articulating this worry further.

changes that we introduce into our environments, that alter our environments, feedback into ourselves as we are influenced, affected, and shaped by that environment” (Strate 2017). Digital media reflects a set of technologies and information-processing systems that work over discrete and quantified packaged bits of binary code. And, as Ott notes, “those features reflect the underlying ‘logic’ of that media form” (2023: 3). He continues: “Habitual use of a media form creates an appetite for that logic. When a media form comes to dominate our social and cultural environment, so too does the underlying logic of that media form” (ibid.). Here, we circle back to the nature of machine architectures that we outlined in Sect. (3.1). Machine architectures are built on microprocessors, which, like all technologies, are neither neutral nor inert. They are manufactured and structured in particular ways that have inherent structural biases toward specific types of information or data. Ott notes, “Repeated exposure to those biases alters how we communicate and interact, how we engage with and respond to events, and even how we make sense of ourselves and our world. In short, our prevailing technologies of communication endlessly and fundamentally remake us in their image” (2023: 1). Ott helpfully identifies three structural biases that have manifested in the ‘digital mind’: intransigence, impertinence, and impulsivity. There are likely more, but for our purposes, these three should outline the wider point.

More specifically, the triad is structured as follows. Computers prioritize and optimize for digitization, execution, and efficiency, and these result in the phenomenological habits of intransigence, impertinence, and impulsivity. Intransigent thought is thought that is dichotomous and dogmatic; impertinent thought is thought that is increasingly insensitive and unresponsive to endlessly evolving contexts; impulsive thought is affective thought. This latter mode is not to disparage the role of emotion and affect in facilitating moral and conscientious deliberations but rather shows how engagement with digital technologies leads to a kind of impulsivity that underdevelops the kind of critical thinking skills required to be context sensitive and respond accordingly. The details do not concern us herein, but a summary of the implications helps to round out the argument of the RAP:

Whereas analogue electronic media were defined by the structural properties of a continuous signal, post-modern fragmentation, and niche messaging, which reflect and promote the logics of contiguity, perspectivism, and deliberateness, digital electronic media are characterized by the structural logics of digitization (binary code), algorithmic execution (input/output), and efficiency (machine logic), which foster modes of thought characterized by intransigence, impertinence, and impulsivity. (2023: 11)

When read through the context of alignment, computer scientists and AI researchers have premised their research on constructing a series of capabilities that include reasoning, knowledge representation, perception, machine vision, and affect recognition, to name a few. There is an odd irony here. The brain is often used as a starting point for inspiration on how to proceed with AI research. But the similarities are only apparent. More pointedly, machine logic is not biology and so the quest is not to make machines more like biology but to overcome the disparity by building machines that are human compatible. Read like this, our AI revolution is marked less by AI becoming more human, but in a manner reminiscent of Thrun and Schmidt’s vision for AI–human interactions, rewiring humanity in fundamental ways that put into question the ‘human compatibility’ that often animates the discussion. Stated as a question, is AI extending our senses, or are we becoming extensions of these AI systems? And, because this technology is not (currently) autonomous from the ones in power of creating it, are we the products of their technology?

The preceding discussion harmonizes well with the notion of the reverse Turing test explored in Frischmann and Selinger (2018). According to them, “We’re not interested in identifying machines engineered to be intelligent so much as we’re interested in identifying humans engineered to be unintelligent” (2018: 184). This is an intriguing and prescient line of thinking, so I will follow it for a bit to identify its relation to the RAP. Frischmann and Selinger are less concerned with machines mimicking humans and more with whether environments can be designed to make humans indistinguishable from machines. Their focus shifts from AI to techno-social engineering (their term), i.e., the structure of human behavior through technological systems that nudge, constrain, or automate decision-making. They offer a series of thought experiments and empirical scenarios designed to examine whether humans are increasingly engineered to ‘not think’—that is, to behave in ways that are devoid of core cognitive capacities such as rationality, creativity, and common sense. Through a series of thought experiments, they invite the reader to consider not only the epistemic or classificatory consequences of this re-engineering (e.g., can we tell humans and machines apart?) but the normative dimension: what does it mean when humans ‘pass’ as machines in these contexts? Their analysis is deeply intertwined with the question of autonomy, dignity, and ‘constructive environments’—settings designed not merely to elicit machine-like behavior in humans temporarily but over longer time-scales: raising concerns about dehumanization and erosion of agency. In an intriguing, if worrying, exploration of this idea, self-driving car companies like Uber have proposed precisely this idea of a constructive environment in situ. So-called smart cities are premised on notoriously hard to predict pedestrian behavior becoming more predictable for

self-driving cars: a managed, curated environment ideal for them, free from the spontaneity and unpredictable elements of human behavior that currently causes major headaches for self-driving car companies (Mitchell 2020).

The RAP is an intellectual continuation of this technological engineering but with a distinct orientation and critical vocabulary. It concerns the way human subjectivity and social interactions are restructured, repackaged, and repurposed to further the dubious quest of creating AGI. Reverse alignment is hard to pin down precisely because the ‘move fast and break things’ mentality that has guided this technology for so long has consequences so diffuse and subtle beyond the testimonials that make the headlines. The RAP concerns a surreptitious assimilation of human subjectivity to the modes and logics of machine operations under the pretense of striking beneficial alignment. While there are valuable voices in the field focusing on *sociotechnical* alignment—that is, alignment that asks whether internally or locally aligned AI systems necessarily align with ideals of human well-being, flourishing, and societies—the vast majority of researchers in this area do not seem particularly concerned with *this* year, *this* context, *this* worrying political trend, *this* testimony of people negatively impacted by their technologies: it is hard to be concerned with the near term when you fully believe the technology you are creating is the dawn of a ‘new era for humanity’ (Zuckerberg; quoted in Carr 2025).

There is a profound worry that the creators of this technology—with their eyes fixated on a horizon that those concerned with near-term consequences do not, supposedly, seem to appreciate—stand in a similar position to Mrs. Jellyby in the Dickens book, *Bleak House*, an affluent philanthropist who donated much of her income to charitable causes far away: “she was a pretty, very diminutive, plump woman of from forty to fifty, with handsome eyes, though they had a curious habit of *seeming to look a long way off*. As if... they could see nothing nearer than Africa” (2003: 44). So concerned with the suffering beyond her own shores, her children, and house fall, as the titular description suggests, into disrepair. Perhaps it is in such a bleak house that current technologies are being developed. Perhaps Sam Altman’s jab “i am a stochastic parrot and so r u” is more a convenient reality than a psychological truth. Perhaps rendering people more predictable is easier than making machines complex beyond ‘scaling laws’. It is, in any case, certainly more lucrative—and Mrs. Jellyby was, if anything, a woman of money.

5 Conclusion

“Technology can make us forget what we know about life”

—Sherry Turkle, *The Empathy Diaries* (2021: Xii)

On a Monday afternoon in August, Jim Acosta—survivor of the horrific Parkland shooting at Marjory Stoneman Douglas High School—sits down with Joaquin Oliver—also a student of the same high school—for an interview. Except the thing is that Joaquin, tragically, was not a survivor of that mass shooting. Instead, he was being ‘vivified’ and animated through stilted Generative AI movements and speaking with Joaquin’s voice modeled on bits of his online writing and home-video footage. The animation is stiff. The cadence of Joaquin’s voice is inconsistent and uncanny. And yet Jim Acosta, a noted political activist and CNN personality, is fully bought in on the significance of this interview to advocate for gun regulation. He engages with, and responds to, Oliver as if this were an entirely normal interview, with an entirely normal interlocutor, who was entirely (and consensually) engaged in the whole affair.²¹

If this sounds somewhat surreal and horrifying, that is because it is. Media critic Parker Molloy called it “turning a murdered child into content”; Charlie Warzel decries “the tech companies that now offer a monkey’s paw in the form of products that can reanimate the dead”; and, to add to matters, Joaquin Oliver’s father is quoted as saying, “Any other Silicon Valley tech guy will say, ‘This is just the beginning’”. But is it the beginning? And the beginning of what? I want to conclude this paper with three observations about the consequences of the RAP. One is social; one is theoretical; one is environmental.

The first is to add to the chorus of people who are collectively identifying a “societal race toward a future that feels bloodless, hastily conceived, and shruggingly accepted” (Warzel 2025). But the race towards this future began with the inculcation and encouragement of transforming ourselves like beings from Pygmalion into ‘digital subjects’—‘subjectivities’ that rely on our ‘data’. “Ultimately”, Kate Crawford writes (2021: 113), “‘data’ has become a bloodless word; it disguises both its material origins and its ends”. This data is liberated from its life in the social—“no longer inconveniently encumbered by moral reasoning, politics, social norms, rights, values, relationships, feelings, contexts, and situations” (Zuboff 2019: 211). It is a shift and transformation to an understanding of others not as finite, embodied, vulnerable, fault-prone beings but towards a transcendent imitation of who we are. In the uncomfortable case of Acosta and the deceased Oliver, it is important to keep in mind that Oliver’s ‘digital twin’ is the product of training data scraped from his online past. Anything that the generated version produces is a product of what the living Oliver has said. Yet

²¹ This event is beautifully if tragically captured in the writing of Charlie Warzel of *The Atlantic* (2025: <https://www.theatlantic.com/technology/archive/2025/08/ai-mass-delusion-event/683909/>).

the interview is intended to elicit a post hoc rumination on the consequences of his slaying. But recall that computers depend on algorithmic execution, on formalized axiomatic principles that enable them to draw their inferences. As Fazi notes, “From this perspective, computer programs, just like axioms, return to us only what is already known. They provide an output that is already implicit in the input” (2018: 4). While Generative AI might produce novel ‘content’, this is best seen as ‘remixed’ ideas—perhaps containing updated information about the events of the Parkland shooting (Oliver did not live through the event so, of course, his digital twin can only know what ‘it’ is taught through online training data). This introduces profound ethical worries. If one argues that the remixed responses generated by Oliver represent his ideas, that stretches the boundaries of consent and conscious endorsement—especially if it concerns a topic and event that happened *after* the person has deceased. As it happens, this is a strategic advantage of datafied selves: “If data is seen as abstract and immaterial, then it more easily falls outside of traditional understandings and responsibilities of care, consent, or risk” (Crawford 2021: 211). The first consequence of the RAP is that we are adapting ourselves, or being made to adapt to, this bloodless future and understanding ourselves through the transformations this digitization encourages.

The second consequence of the RAP is that, especially with the Generative AI revolution, we risk establishing a scientific monoculture in machine learning and AI research. Because of the massive gold rush involved in Generative AI technologies, most companies and research institutes have suspended alternative approaches to AI. OpenAI used to have a robotics department, if you can believe it; and after the launch of ChatGPT, Google DeepMind rerouted funding and resources for other projects towards catching up with OpenAI (Hao 2025). This relates back to the third observation of the RAP that I outlined in the beginning: namely that AI might not ever need to become more complex to the value landscape because we may have adapted ourselves to a hastily constructed environment that is more manageable. This is a subtle, but significant departure from arguments found in Dreyfus and Searle, which, in effect, say that AI could never express the capacities and interiority associated with human subjectivity. There were, and still are, myriad other roads through which AI and perhaps AGI could be achieved—in conjunction with LLMs. Embodied and social robotics, soft robotics, alternative computing paradigms, hardware innovations, and myriad other research avenues (Harrison et al 2022) become systematically closed off as resources are funneled into the most financially lucrative short-term investment. Alternative computing paradigms are marginalized as the morass of large language models and the epistemic distortions they introduce are promised as the basis from which AGI

might spontaneously emerge. It is possible to envision a future involving a kind of synthesis between mainstream deep-learning approaches and robotics, for instance, but at the moment, this avenue seems unlikely due to the gravitational pull of LLMs. Alternative computing paradigms could, in tandem with LLMs, contribute to further ways in which we can more accurately represent values without relying on the explicitness of industry-standard practices that lend themselves easily and readily to value capture. The RAP, in articulating a trend in a particular way, tries to shake the conviction that the path forward for AI is the one envisioned by the people who have vested interest in LLM being the ‘unlock’ for AGI—people who, in many cases, we should not have high confidence in but who nonetheless shape our futures.

Lastly, it would be remiss of me to introduce the RAP without talking about the very real material and environmental impact these technologies are having around the globe. Tar lakes in Inner Mongolia; devastated landscapes in Indonesia; abandoned, monstrous holes in the Chilean Andes: the drive for ever bigger, ever stronger, ever more voracious LLMs is transforming the very environment without which the entire Generative AI revolution would not be possible. This is what makes the modern AI industry a kind of planetary computational network, reliant on way stations and outposts distributed across the globe and often out of sight of those who develop the technology: usually at the peripheries and to the detriment of underdeveloped, impoverished places, involving exploited and low pay laborers who are *actually* “feeling the AGI”, as Ilya Sutskever likes to put it. While it goes beyond the scope of this paper to identify how all these are connected, the RAP identifies the numerous ways in which the loci of alignment are distributed and negotiated, suggesting that striking beneficial alignment requires a social sciences lens typically ignored by the ones designing these systems.

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Declarations

Conflict of interest The authors declare no competing interests.

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References

- Agarwal D, Naaman M, Vashistha A (2025) AI suggestions homogenise writing toward western styles and diminish cultural nuances. arXiv.org: <https://arxiv.org/abs/2409.11360>
- Amartya S (1973) Behavior and the concept of preference. *Economica* 40(159):241–259
- Anderson E (2001) Unstrapping the straightjacket of ‘preference’: a comment on Amartya Sen’s contributions to philosophy and economics. *Econ Philos* 17(1):21–38
- Anderson M, Anderson L (2007) The status of machine ethics: a report from the AAAI Symposium. *Minds Mach* 17:1–10
- Arendt H (1998) *The human condition*. University of Chicago Press, Chicago, Illinois
- Barrett LF, Adolphs R, Marsella S, Martinez AM, Pollak SD (2019) Emotional expressions reconsidered: challenges to inferring emotion from human facial movements. *Psychol Sci Public Interest* 20(1):1–68
- Bender E, Hanna A (2025) *The AI Con: How to Fight Big Tech's Hype and Create the Future We Want*. Penguin Random House.
- Bender E, Koller A (2020) Climbing towards NLU: on meaning, form, and understanding in the age of data. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 5185–5198.
- Bender E, McMillan-Major A, Schmittell S, Gebru T (2021) On the dangers of stochastic parrots: can language models be too big?. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 610–623.
- Birhane A (2021) The impossibility of automating ambiguity. *Artif Life* 27(1):44–61
- Bostrom N (2014) *Superintelligence: paths, dangers, strategies*. Oxford University Press, Oxford, United Kingdom
- Boyd D (2024) The structuring work of algorithms. *Daedalus* 152(1):236–240
- Bradford A (2023) *Digital Empires: The Global Battle to Regulate Technology*. Oxford: Oxford University Press.
- Buolamwini J (2024) *Unmasking AI: my mission to protect what is human in a world of machines*. Random House, London
- Campolo A, Crawford K (2020) Enchanted determinism: power without responsibility in artificial intelligence. *Engag Sci Technol Soc* 6(2020):1–19
- Carr N (2025) *Superbloom: how technologies of connection tear us apart*. Norton & Company, New York, N.Y.
- Chayka K (2024) *Filter World: How Algorithms Flattened Culture*. Penguin Random House: DoubleDay.
- Christian B (2020) *The alignment problem: machine learning and human values*. W.W. Norton & Company, New York, New York
- Crawford K (2021) *Atlas of AI: power, politics, and the planetary costs of artificial intelligence*. Yale University Press, New Haven, Connecticut
- Doctor T, Witkowski O, Solomonova E, Duana B, Levin M (2022) ‘Biology, Buddhism, and AI: care as the driver of intelligence. *Entropy* 24(5):710
- Edelman J, Zhi-Xuan T, Lowe R, Klingefjord O et al (2025) Full-stack alignment: co-aligning AI and institutions with thick models of value. <https://www.full-stack-alignment.ai/paper>
- Ekman P, Friesen W (1978) *Facial action coding system: a technique for the measurement of facial movement*. Consulting Psychologists Press, Palo Alto
- Foster J (2023) From thin to thick: toward a politics of human-compatible AI. *Public Cult* 35:3
- Frischmann B, Selinger E (2018) *Re-engineering humanity*. Cambridge University Press, Cambridge
- Green BP (2019) Artificial intelligence, decision making, and moral deskilling. markulla center for applied ethics, Santa Clara University: <https://www.scu.edu/ethics/focus-areas/technology-ethics/resources/artificial-intelligence-decision-making-and-moral-deskilling/>
- Hao K (2025) *Empire of AI: inside the reckless race for total domination*. Penguin Random House, London, United Kingdom
- Harrison D, Rorot W, Laukaityte U (2022) Mind the matter: active matter, soft robotics, and the making of bio-inspired artificial intelligence. *Front Neurorobot*. <https://doi.org/10.3389/fnbot.2022.880724/full>
- Hubinger E, van Merwijk C, Mikulik V, Skalse J, Garrabrant S (2019) Risks from learned optimisation in advanced machine learning systems. arXiv.org: <https://arxiv.org/abs/1906.01820>
- Kazienko P, Cambria E (2024) Toward responsible recommender systems. *IEEE Intell Syst* 39(3):5–12. <https://doi.org/10.1109/MIS.2024.3398190>
- Klingefjord O, Lowe R, Edelman J (2024) What are human values, and how do we align AI to them?. arXiv preprint [arXiv:2401.12358](https://arxiv.org/abs/2401.12358).
- Koch B, Denton E, Hanna A, Foster J (2021) Reduced, reused, and recycled: the life of a dataset in a machine learning research. 35th Conference on Neural Information Processing Systems, Sydney, Australia. <https://arxiv.org/abs/2112.01716>
- Korte M (2020) The impact of the digital revolution on human brain and behavior: where do we stand? *Dialogues Clin Neurosci* 22:101–111
- Krakovna V (2018) Specification gaming examples in AI. Word Press: <https://vkrakovna.wordpress.com/2018/04/02/specification-gaming-examples-in-ai/>
- Kulveit J, Douglas R, Ammann N, Turan D, Krueger D, Duvenaud, D (2025) Gradual disempowerment: systemic existential risks from incremental AI development. arXiv: <https://arxiv.org/abs/2501.16946>
- Lake BM, Ullman TD, Tenenbaum JB, Gershman SJ (2016) Building machines that learn and think like people. *Behav Brain Sci* 40:e253
- Merry SE (2016) *The seductions of quantification: measuring human rights, gender violence, and sex trafficking*. The University of Chicago Press, Chicago, Illinois
- Millière R, Buckner C (2024) A Philosophical Introduction to Language Models--Part I: Continuity With Classic Debates. arXiv: <https://doi.org/10.48550/arXiv.2401.03910>
- Millière R (2025) Normative conflicts and shallow alignment. *Philos Stud* 182:2035–2078
- Mitchell M (2020) *Artificial intelligence: a guide for thinking humans*. Picador, New York, New York

- Moore C, Gill C, Bliss N, Butler K, Forrest S, Lopresti D, Maher M, Mentis H, Shekhar S, Stent A, Turk M (2025) 'Concerning the Responsible Use of AI in the U.S. Criminal Justice System'. *Communications of the ACM*, vol. 68(9):41–44.
- Narayanan A, Kapoor S (2025) AI snake oil: what AI can do, what it can't, and how to tell the difference. Princeton University Press, Princeton, N.J.
- Ngo R, Chan L, Mindermann S (2025) The alignment problem from a deep learning perspective. arXiv: <https://arxiv.org/abs/2209.00626>
- Nguyen CT (2024) Value capture. *J Ethics Soc Phil* 27:469
- Ott B (2023) The digital mind: how computers (re)structure human consciousness. *Philosophies* 8:4
- Pellegrino A, Stasi A (2024) A bibliometric analysis of the impact of media manipulation on adolescent mental health: Policy recommendations for algorithmic transparency. *Online J Commun Media Technol* 14(4):e202453
- Penn J (2020) Inventing intelligence: on the history of complex information processing and artificial intelligence in the united states in the mid-twentieth century. Apollo—University of Cambridge Repository.
- Porter TM (2020) *The Rise of Statistical Thinking: 1820–1900*. Princeton University Press, Princeton, New Jersey
- Raji ID, Bender EM, Paullada A, Denton E, Hanna A (2021) AI and the everything in the whole wide world benchmark. arXiv: <https://doi.org/10.48550/arXiv.2111.15366>
- Russel S (2019) *Human compatible: Artificial Intelligence and the problem of control*. Penguin Publishing, New York, New York
- Scott JC (1998) *Seeing like a state: How certain schemes to improve the human condition have failed*. Yale University Press, New Haven, Connecticut
- Seth A (2025) Conscious artificial intelligence and biological naturalism. *Behav Brain Sci*. <https://doi.org/10.1017/S0140525X25000032>
- Strate L (2017) *Media ecology: an approach to understanding the human condition*. Peter Lang, New York, N.Y.
- Tegmark M (2018) *Life 3.0: being human in the age of artificial intelligence*. Penguin Books, London, United Kingdom
- Turk V (2025) *The Great Language Flattening*. The Atlantic: <https://www.theatlantic.com/technology/archive/2025/04/great-language-flattening/682627/>
- Turkle S (2021) *The Empathy Diaries*. Penguin Publishing Group. New York: New York.
- Vallor S (2015) Moral deskilling and upskilling in a new machine age: reflections on the ambiguous future of character. *Philos Technol* 28:107–124
- Vicario MD, Vivaldo G, Bessi A, Zollo F, Scala A, Caldarelli G, Quattrociocchi W (2016) Echo chambers: emotional contagion and group polarization on facebook. *Sci Rep* 6(1):37825
- Warzel D (2025) AI Is A Mass Delusion Event. *The Atlantic*. <https://www.theatlantic.com/technology/archive/2025/08/ai-mass-delusion-event/683909/>
- Winner L (1988) *The whale and the reactor: a search for limits in an age of high technology*, 2nd edn. Chicago University Press, Chicago, Illinois
- Yudkowsky E, Soares N (2025) *If Anyone Builds It, Everyone Dies*. Little Brown Publishers. New York
- Zhi-Xuan T, Carroll M, Franklin M, Ashton H (2024) Beyond preferences in AI alignment. *Philos Stud*. <https://doi.org/10.1007/s11098-024-02249-w>
- Zuboff S (2019) *Surveillance capitalism: the fight for a human future at the new frontier of power*. Public Affairs, London, United Kingdom

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